

The self-auditing risk layer for Alpaca options.

An autonomous agent that trades defined-risk spreads through Alpaca's CLI — and re-prices the trades it refused, to prove the refusals were right.

20,000-path Monte Carlo VaR · Counterfactual refusal scoring

ACCOUNT

PA3BY0HJORNC

fresh, \$100,000, options level 3

EQUITY

\$100,149.93

live paper trading

EXECUTION

Alpaca CLI

multi-leg, on a schedule

THE PROBLEM

Risk paralysis.

Give a language model a brokerage API and it finds a reason to trade. Put a filter in front of it and it finds a reason to never trade.

Both failures look identical from the outside — a flat equity curve and a confident explanation. Nobody can tell an agent with discipline from an agent that is simply broken, because nobody measures the trades that were *not* taken.

UNCALIBRATED FILTER

0 trades

looks like discipline

NO FILTER

blown account

looks like conviction

WHAT'S MISSING

the counterfactual

what the refusals would have done

Every refusal becomes a testable claim.

01

Refuse

The gate vetoes a spread and logs its legs, credit and reason to an append-only ledger.

02

Re-price

Those exact legs are re-priced later at live marks — what the refused trade would have earned.

03

Grade

Scored symmetrically. A refusal that avoided a loss counts; so does one that missed a profit.

04

Recalibrate

The threshold moves on the evidence, and the scorer keeps grading the new one.

Before execution, a 20,000-path Monte Carlo prices the distribution — probability of profit, 95% VaR, expected shortfall. A trade with negative expected value is refused however much premium it appears to pay.

Predicted before the trade. Verifiable after it.

	MONTE CARLO, BEFORE	LIVE ALPACA EXECUTION
P&L on the 17-lot	+\$145 expected	+\$153 realised
Probability of profit	92%	won
95% VaR	-\$1,428	not breached
Structural max loss	-\$1,428	0 of 20,000 paths exceeded

open cd44d308-8609-4044-8994-3137d255fc78 17 @
-0.16 2026-09-01T17:10:59Z
close f67cd64a-a82c-4db8-a877-52de37037761 17 @
+0.07 2026-09-02T14:57:24Z

And the honest half: all four gate vetoes would have been profitable. We published that unchanged and lowered the floor on the evidence.

Alpaca options production hardening.

Three undocumented behaviours, each found by running against the live API, each carrying a workaround any team building options on Alpaca will need.

- **Multi-leg limit sign is inverted.** A positive net limit is read as a *debit* — it silently authorises paying to enter a credit spread.
- **Greeks require --feed indicative.** The default feed 403s without an OPRA agreement; snapshots return nulls and delta targeting dies quietly.
- **Intraday portfolio history is wrong in paper.** Reports exactly \$100,000 too high.

MIT licensed. The workaround layer is the contribution: it turns Alpaca's options API from something you discover by trial into something you build on.